MSC 201 **23774**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Twenty items have sample mean mass 5 g. What is their total mass?	A. 5 g. B. 100 g. C. 0.25 g. D. 25 g. 10 pts
2. The plus-four estimate adds two successes and two failures. With 8 successes out of 40, what adjusted fraction is used?	A. 8/44. B. 10/44. C. 12/44. D. 10/40. 10 pts
3. A rate estimate rises from 40% to 45%. What is its increase in percentage points?	A. 5 percentage points. B. 0.05 percentage points. C. 45 percentage points. D. 12.5 percentage points. 10 pts
4. The population mean is unknown. Why are repeated-sample models useful for estimation?	A. They guarantee every sample equals the population. B. They describe how far a sample mean might vary around that population mean. C. They make the population mean change randomly every day. D. They remove the need for representative sampling. 10 pts
5. Why does a t interval use a larger critical multiplier than z at small degrees of freedom?	A. The confidence level is automatically 100%. B. Estimating SD adds uncertainty reflected in heavier tails. C. The observations must all be larger. D. The population mean is random in the formula. 10 pts
6. Population mean is 20; SE of sample means is 2. An observed sample mean is 24. What standardized distance does it have?	A. 0.5. B. 12. C. 4. D. 2. 10 pts

Test continues on the other side of the sheet.

MSC 201 **23774**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. After a specific interval is computed, what is correct about its relation to a fixed population mean?	A. The endpoints must equal the smallest and largest observation. B. It either contains that mean or does not; 95% describes the procedure. C. The fixed mean changes to fit it with probability 95%. D. The mean occupies 95% of the interval. 10 pts
8. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It stays unchanged. B. It increases. C. It decreases. D. It becomes zero. 10 pts
9. For independent observations with finite variance, compare repeated means at n=1 and n=100. What is generally expected?	A. The n=1 histogram must be normal. B. The population itself becomes less spread. C. Means at n=100 are less spread and often more nearly normal. D. Every size-100 mean is identical. 10 pts
10. Why represent a single binomial count by an interval when using a normal approximation?	A. The count can take fractional values. B. An interval has area under a continuous curve, while a single point has zero area. C. A count always equals its expected value. D. The normal distribution has positive probability at every exact point. 10 pts

23774

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

B.

10 pts

9.

C.

10 pts

10.

B.

10 pts

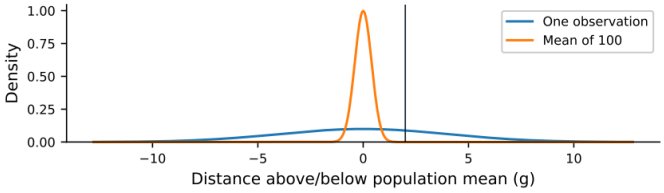
MSC 201 **60897**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. For a normal population, compare one mass exceeding mean+2 g with the mean of 100 independent masses exceeding the same cutoff. Which is more likely? 	A. The 100-mass mean event. B. The individual mass event. C. They are equally likely. D. Both have probability zero. 10 pts
2. For a normal approximation to the probability of at least a given count, what should happen to the interval representing the cutoff count?	A. Include the entire interval and all intervals above it. B. Exclude the interval and include only those above it. C. Use just a single point at the cutoff. D. Include only the intervals below it. 10 pts
3. From the same sample, a 90% and 99% mean interval are built with the same method. Which statement fits?	A. The 90% interval is wider. B. The 99% interval has a different sample mean. C. Both widths must be identical. D. The 99% interval is wider but has the same center. 10 pts
4. Why report a confidence interval instead of only a sample mean?	A. It guarantees the true mean is the midpoint. B. It lists every individual observation. C. It removes sampling bias. D. It expresses uncertainty in estimating the population mean. 10 pts
5. A sample contains 100 observations, but population SD is unknown. Does large n make that SD known?	A. Yes; the sample SD becomes the population parameter by definition. B. No; it is still estimated from data. C. Yes; n>=30 makes population SD exact. D. No; the sample mean must also be unknown. 10 pts

Test continues on the other side of the sheet.

MSC 201 **60897**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

6. A histogram is made from 500 means, each based on 20 observations. How many values are in that histogram?	A. 10,000. B. 520. C. 500. D. 20. 10 pts
7. For independent observations with finite variance, compare repeated means at n=1 and n=100. What is generally expected?	A. Means at n=100 are less spread and often more nearly normal. B. The population itself becomes less spread. C. Every size-100 mean is identical. D. The n=1 histogram must be normal. 10 pts
8. The plus-four estimate adds two successes and two failures. With 8 successes out of 40, what adjusted fraction is used?	A. 10/40. B. 8/44. C. 10/44. D. 12/44. 10 pts
9. 40 of 100 sampled jobs succeed. What is the sample success proportion?	A. 0.60. B. 0.40. C. 40. D. 2.5. 10 pts
10. The number of independent items rises from 25 to 100. What happens to expected sum and SD of sum?	A. Both quadruple. B. Expected sum stays fixed; SD halves. C. Both halve. D. Expected sum quadruples; SD doubles. 10 pts

60897

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

B.

10 pts

TEST PAGE 2

6.

C.

10 pts

7. A.

10 pts

8.

C.

10 pts

9.

B.

10 pts

10.

D.

10 pts

MSC 201 **61330**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Individual masses have SD 20 g; means of 25 independent masses have SD 4 g. Which spread applies to one package?	A. 16 g. B. 20 g. C. 4 g. D. 0.8 g. 10 pts
2. Using the same sample and interval method, confidence increases. What happens to the proportion margin?	A. It increases. B. The sample proportion becomes the confidence level. C. It stays unchanged. D. It decreases. 10 pts
3. For independent items, the SD of a sum is sqrt(n) times individual SD. Why is n times SD generally wrong?	A. The total is always fixed. B. Means cannot be added. C. All individual deviations have the same sign. D. Independent variances add; SD is then a square root. 10 pts
4. A 95% confidence interval estimates a population use rate. Does 95% mean that 95% of the population uses the service?	A. No; it means 95 people were surveyed. B. Yes; all sample proportions equal 0.95. C. Yes; confidence equals the use rate. D. No; 95% describes the interval procedure's coverage. 10 pts
5. A symmetric interval is 8 to 12. What is the margin of error?	A. 4. B. 2. C. 12. D. 10. 10 pts
6. SE is 4 at n=25. What is SE at n=100, with the same population SD and independence?	A. 4. B. 1. C. 2. D. 8. 10 pts

Test continues on the other side of the sheet.

MSC 201 **61330**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Why represent a single binomial count by an interval when using a normal approximation?	A. A count always equals its expected value. B. An interval has area under a continuous curve, while a single point has zero area. C. The normal distribution has positive probability at every exact point. D. The count can take fractional values. 10 pts
8. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 256. B. 15. C. 16. D. 17. 10 pts
9. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. The true sample mean can actually be below every observation. B. The normal model is exact at every tail. C. The observations must include negative waits. D. It is approximation error outside the true support. 10 pts
10. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Sampling quality and independence. B. Whether the mean is a whole number. C. Whether every observation equals the mean. D. Whether the interval includes zero. 10 pts

61330

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

B.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

B.

10 pts

9.

D.

10 pts

10. A.

10 pts

MSC 201 **82427**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A 95% confidence interval estimates a population use rate. Does 95% mean that 95% of the population uses the service?	A. No; it means 95 people were surveyed. B. Yes; confidence equals the use rate. C. No; 95% describes the interval procedure's coverage. D. Yes; all sample proportions equal 0.95. 10 pts
2. Sample SD is 6 minutes and n=9; sqrt(9)=3. What estimated SE is used?	A. 18 minutes. B. 3 minutes squared. C. 6 minutes. D. 2 minutes. 10 pts
3. A mean interval is 10 to 14 minutes. Which report matches its target?	A. Exactly 95 of the next 100 durations will lie there. B. Every future duration will be between 10 and 14. C. The procedure estimates the population mean duration between 10 and 14 minutes. D. The median is guaranteed to equal 12. 10 pts
4. SE is 4 at n=25. What is SE at n=100, with the same population SD and independence?	A. 1. B. 4. C. 8. D. 2. 10 pts
5. The CLT supports an approximately normal distribution of large-sample means. What does it not claim?	A. Larger independent samples can have smaller SE. B. Means can have approximately normal sampling distributions. C. Individual observations become normally distributed. D. Sampling distributions concern statistics. 10 pts
6. For a normal approximation to the probability of at least a given count, what should happen to the interval representing the cutoff count?	A. Use just a single point at the cutoff. B. Include only the intervals below it. C. Exclude the interval and include only those above it. D. Include the entire interval and all intervals above it. 10 pts

Test continues on the other side of the sheet.

MSC 201 **82427**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A sample has 2 successes and 98 failures. Does a large total alone justify the ordinary normal-approximation proportion interval?	A. Yes; failures are numerous. B. Yes; n=100 guarantees every condition. C. No; the observed success count is too sparse. D. No; proportions cannot be estimated at all. 10 pts
8. A sum distribution has mean 100 and SD 4. What total is 2 SD above its mean?	A. 102. B. 104. C. 108. D. 200. 10 pts
9. In 1,000 repetitions, each sample contains 25 independent observations. What n belongs in SD/sqrt(n) for the mean?	A. 25,000. B. 1,000. C. 25. D. 975. 10 pts
10. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Whether the interval includes zero. B. Sampling quality and independence. C. Whether every observation equals the mean. D. Whether the mean is a whole number. 10 pts

82427

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5.

C.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

C.

10 pts

9.

C.

10 pts

10.

B.

10 pts

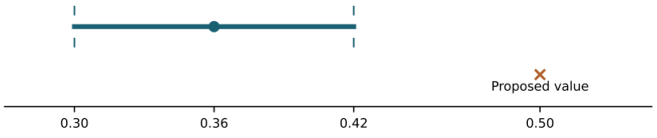
MSC 201 **67316**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Independent sample size quadruples with population SD fixed. What happens to SE of the mean?	A. It stays unchanged. B. It quarters. C. It doubles. D. It halves. 10 pts
2. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. The observations must include negative waits. B. The normal model is exact at every tail. C. It is approximation error outside the true support. D. The true sample mean can actually be below every observation. 10 pts
3. A population proportion interval is 0.30 to 0.42. How should a proposed proportion 0.50 be described? 	A. Proved impossible under every future sample. B. Compatible because 0.50 is a valid probability. C. The required sample estimate. D. Outside the interval and incompatible with this interval estimate. 10 pts
4. A sample contains 100 observations, but population SD is unknown. Does large n make that SD known?	A. Yes; the sample SD becomes the population parameter by definition. B. Yes; n>=30 makes population SD exact. C. No; it is still estimated from data. D. No; the sample mean must also be unknown. 10 pts
5. An open voluntary poll has 50,000 replies and a very small calculated margin. What remains uncertain?	A. Whether 50,000 exceeds 30. B. Whether its respondents represent the target population. C. Whether a proportion is numerical. D. Whether the calculated margin must be negative. 10 pts
6. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It becomes zero. B. It increases. C. It decreases. D. It stays unchanged. 10 pts

Test continues on the other side of the sheet.

MSC 201 **67316**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A population has mean 50. Where are unbiased random-sample means centered?	A. At 50 divided by sample size. B. At zero. C. At the population SD. D. At 50. 10 pts
8. For a normal approximation to the probability of at most a given count, which intervals belong in the shaded area?	A. Only the intervals below the cutoff count's interval. B. Only the cutoff count's interval. C. The cutoff count's interval and all intervals below it. D. Only the intervals above the cutoff count's interval. 10 pts
9. A 95% mean interval is 10 to 14 minutes. Does this say 95% of individual durations lie there?	A. Yes; that is the meaning of any confidence interval. B. No; it estimates the population mean. C. Yes; the mean and individuals have identical spread. D. No; it instead guarantees every duration lies there. 10 pts
10. All items in a batch tend to be heavy or light together. What is questionable about sqrt(n) times individual SD for the total?	A. It retains mass units. B. It uses the sample size. C. It assumes independent variation. D. It predicts a nonnegative SD. 10 pts

67316

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

B.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

C.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **18524**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A rate estimate rises from 40% to 45%. What is its increase in percentage points?	A. 45 percentage points. B. 0.05 percentage points. C. 5 percentage points. D. 12.5 percentage points. 10 pts
2. Fifty readings taken seconds apart from one fluctuating machine are treated as independent. What should be checked?	A. Whether readings have units. B. Dependence between nearby readings. C. Whether 50 is an integer. D. Whether the mean can be calculated. 10 pts
3. Twenty items have sample mean mass 5 g. What is their total mass?	A. 25 g. B. 5 g. C. 0.25 g. D. 100 g. 10 pts
4. A known-SD mean interval currently has margin 1. To target margin 0.5 at the same confidence, how should n change?	A. Add 1 to n. B. Halve n. C. Multiply n by 4. D. Multiply n by 2. 10 pts
5. After a specific interval is computed, what is correct about its relation to a fixed population mean?	A. The mean occupies 95% of the interval. B. The fixed mean changes to fit it with probability 95%. C. The endpoints must equal the smallest and largest observation. D. It either contains that mean or does not; 95% describes the procedure. 10 pts
6. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A prediction interval for a single future observation. B. A known-SD z interval treating s as known population SD. C. A proportion interval regardless of variable type. D. A one-sample t interval. 10 pts

Test continues on the other side of the sheet.

MSC 201 **18524**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A question asks whether the combined duration of 40 tasks exceeds one hour. Which variable should be modeled?	A. The median alone. B. The count of tasks alone. C. The sum of 40 durations. D. One individual duration. 10 pts
8. Larger independent samples yield less variable means. Does this make individual packages less variable?	A. Only if more samples are plotted. B. Yes; every package moves toward the mean. C. No; population spread stays the same. D. Yes; individual SD becomes zero. 10 pts
9. 400 people are sampled without replacement from 20,000. Does this meet the at-most-10% sample-fraction guideline?	A. No; n must be below 30. B. Yes; the fraction is 2%. C. No; the fraction is 20%. D. Yes; any random sample meets it. 10 pts
10. To halve SE, how must an independent sample's size change?	A. Add 4 regardless of its current size. B. Multiply n by 2. C. Multiply n by 4. D. Divide n by 2. 10 pts

18524

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

D.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

C.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **24684**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Larger independent samples yield less variable means. Does this make individual packages less variable?	A. No; population spread stays the same. B. Yes; individual SD becomes zero. C. Yes; every package moves toward the mean. D. Only if more samples are plotted. 10 pts
2. Two samples have the same observed mean above a claimed population mean and the same population SD, but sizes 25 and 100. Which mean is farther above in SE units?	A. The larger sample changes the claimed mean. B. The size-100 mean. C. The size-25 mean. D. They are equally far. 10 pts
3. Everyone in a few selected clubs is surveyed. What challenges treating all responses as independent?	A. Randomly selecting clubs guarantees individual independence. B. Members of the same club may have correlated responses. C. Club members cannot have binary responses. D. The sample proportion cannot be computed. 10 pts
4. A sum event and its equivalent mean event use correctly matched centers and SDs. How do their z-scores compare?	A. The mean z-score is always zero. B. They are equal. C. Their signs must be opposite. D. The sum z-score is n times larger. 10 pts
5. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It increases. B. It becomes zero. C. It decreases. D. It stays unchanged. 10 pts
6. A 95% mean interval is 10 to 14 minutes. Does this say 95% of individual durations lie there?	A. No; it estimates the population mean. B. Yes; that is the meaning of any confidence interval. C. Yes; the mean and individuals have identical spread. D. No; it instead guarantees every duration lies there. 10 pts

Test continues on the other side of the sheet.

MSC 201 **24684**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. The CLT supports an approximately normal distribution of large-sample means. What does it not claim?	A. Sampling distributions concern statistics. B. Individual observations become normally distributed. C. Means can have approximately normal sampling distributions. D. Larger independent samples can have smaller SE. 10 pts
8. An exponential population has mean 5. Which is generally more unusual under independent sampling: one duration above 7 or 100 durations averaging above 7?	A. Neither can occur. B. They have equal probability. C. The average of 100 above 7. D. The single duration above 7. 10 pts
9. The success proportion is 0.35. What is the failure proportion?	A. 0.175. B. 0.35. C. 0.65. D. 1.35. 10 pts
10. A small random independent sample comes from a normal population. Is $n<30$ alone a reason to reject a t interval?	A. No; it must use a proportion interval. B. No; small normal samples can support the t method. C. Yes; t requires at least 30 in every case. D. Yes; normality is irrelevant below 30. 10 pts

24684

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. B.

10 pts

4. B.

10 pts

5. A.

10 pts

6. A.

10 pts

TEST PAGE 2

7. B.

10 pts

8. C.

10 pts

9. C.

10 pts

10. B.

10 pts

MSC 201 **26148**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. After a specific interval is computed, what is correct about its relation to a fixed population mean?	A. The fixed mean changes to fit it with probability 95%. B. The mean occupies 95% of the interval. C. It either contains that mean or does not; 95% describes the procedure. D. The endpoints must equal the smallest and largest observation. 10 pts
2. From the same sample, a 90% and 99% mean interval are built with the same method. Which statement fits?	A. The 99% interval has a different sample mean. B. The 99% interval is wider but has the same center. C. The 90% interval is wider. D. Both widths must be identical. 10 pts
3. Population mean is 20; SE of sample means is 2. An observed sample mean is 24. What standardized distance does it have?	A. 12. B. 2. C. 0.5. D. 4. 10 pts
4. Larger independent samples yield less variable means. Does this make individual packages less variable?	A. Yes; every package moves toward the mean. B. Yes; individual SD becomes zero. C. Only if more samples are plotted. D. No; population spread stays the same. 10 pts
5. A sample of 200 is drawn without replacement from 500. Does it meet the course's at-most-10% independence guideline?	A. No; samples must contain fewer than 10 observations. B. No; it is 40% of the population. C. Yes; 200 is larger than 30. D. Yes; every random sample meets the guideline. 10 pts
6. Sample SD is 6 minutes and n=9; sqrt(9)=3. What estimated SE is used?	A. 6 minutes. B. 18 minutes. C. 3 minutes squared. D. 2 minutes. 10 pts

Test continues on the other side of the sheet.

MSC 201 **26148**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A sum distribution has mean 100 and SD 4. What total is 2 SD above its mean?	A. 102. B. 200. C. 104. D. 108. 10 pts
8. For a comparison holding the observed proportion and confidence fixed, n quadruples. What happens to estimated margin?	A. It quarters. B. It doubles. C. It stays unchanged. D. It halves. 10 pts
9. For a normal approximation to the probability of at most a given count, which intervals belong in the shaded area?	A. Only the cutoff count's interval. B. The cutoff count's interval and all intervals below it. C. Only the intervals below the cutoff count's interval. D. Only the intervals above the cutoff count's interval. 10 pts
10. A rate estimate rises from 40% to 45%. What is its increase in percentage points?	A. 12.5 percentage points. B. 5 percentage points. C. 0.05 percentage points. D. 45 percentage points. 10 pts

26148

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5.

B.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

D.

10 pts

9.

B.

10 pts

10.

B.

10 pts

MSC 201 **55421**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. After a specific interval is computed, what is correct about its relation to a fixed population mean?	A. The fixed mean changes to fit it with probability 95%. B. The mean occupies 95% of the interval. C. It either contains that mean or does not; 95% describes the procedure. D. The endpoints must equal the smallest and largest observation. 10 pts
2. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. The observations must include negative waits. B. It is approximation error outside the true support. C. The normal model is exact at every tail. D. The true sample mean can actually be below every observation. 10 pts
3. Which quantity describes how sample means vary from sample to sample?	A. The sample size. B. The range of all individuals alone. C. Standard error. D. The population mean. 10 pts
4. A symmetric proportion interval is 0.20 to 0.30. What is the estimate and margin?	A. Estimate 0.30; margin 0.25. B. Estimate 0.25; margin 0.05. C. Estimate 0.10; margin 0.05. D. Estimate 0.20; margin 0.10. 10 pts
5. Population SD doubles while sample size stays fixed. What happens to SE?	A. It stays unchanged. B. It doubles. C. It quadruples. D. It halves. 10 pts
6. Why use planning proportion 0.5 when no prior estimate is available?	A. It makes the sample unbiased automatically. B. It guarantees half the population has the trait. C. It maximizes p(1-p), giving a conservative required sample size. D. It minimizes all margins to zero. 10 pts

Test continues on the other side of the sheet.

MSC 201 **55421**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Whether every observation equals the mean. B. Sampling quality and independence. C. Whether the interval includes zero. D. Whether the mean is a whole number. 10 pts
8. For a normal approximation to the probability of at least a given count, what should happen to the interval representing the cutoff count?	A. Use just a single point at the cutoff. B. Include only the intervals below it. C. Exclude the interval and include only those above it. D. Include the entire interval and all intervals above it. 10 pts
9. To estimate a population mean with unknown population SD using a sample SD, which standard method fits?	A. A one-sample t interval. B. A prediction interval for a single future observation. C. A proportion interval regardless of variable type. D. A known-SD z interval treating s as known population SD. 10 pts
10. All items in a batch tend to be heavy or light together. What is questionable about sqrt(n) times individual SD for the total?	A. It predicts a nonnegative SD. B. It retains mass units. C. It uses the sample size. D. It assumes independent variation. 10 pts

55421

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

D.

10 pts

9. A.

10 pts

10.

D.

10 pts

MSC 201 **86543**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Why use planning proportion 0.5 when no prior estimate is available?	A. It makes the sample unbiased automatically. B. It maximizes p(1-p), giving a conservative required sample size. C. It minimizes all margins to zero. D. It guarantees half the population has the trait. 10 pts
2. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. The true sample mean can actually be below every observation. B. The observations must include negative waits. C. The normal model is exact at every tail. D. It is approximation error outside the true support. 10 pts
3. For a normal approximation to the probability of at least a given count, what should happen to the interval representing the cutoff count?	A. Include only the intervals below it. B. Exclude the interval and include only those above it. C. Include the entire interval and all intervals above it. D. Use just a single point at the cutoff. 10 pts
4. Population SD is known and the sample is large. What still needs checking for the usual mean interval?	A. Whether the mean is a whole number. B. Whether the interval includes zero. C. Whether every observation equals the mean. D. Sampling quality and independence. 10 pts
5. For comparison, sample SD and confidence are held fixed while n increases. What happens to a t mean interval's margin?	A. It increases because there are more observations. B. It stays unchanged. C. It becomes the population SD. D. It decreases as SE and the critical t value decrease. 10 pts
6. A rate estimate rises from 40% to 45%. What is its increase in percentage points?	A. 12.5 percentage points. B. 0.05 percentage points. C. 5 percentage points. D. 45 percentage points. 10 pts

Test continues on the other side of the sheet.

MSC 201 **86543**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. In 1,000 repetitions, each sample contains 25 independent observations. What n belongs in SD/sqrt(n) for the mean?	A. 25. B. 975. C. 1,000. D. 25,000. 10 pts
8. Twenty items have sample mean mass 5 g. What is their total mass?	A. 100 g. B. 5 g. C. 25 g. D. 0.25 g. 10 pts
9. Two samples have the same observed mean above a claimed population mean and the same population SD, but sizes 25 and 100. Which mean is farther above in SE units?	A. The larger sample changes the claimed mean. B. The size-100 mean. C. They are equally far. D. The size-25 mean. 10 pts
10. What does a 95% confidence procedure mean in repeated appropriate sampling?	A. The sample mean equals the population mean 95% of the time. B. Every interval has 95% of its length below the mean. C. About 95% of intervals constructed this way contain the fixed population mean in the long run. D. 95% of observations lie in every interval. 10 pts

86543

TEST PAGE 1

1.

B.

10 pts

2.

D.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

C.

10 pts

TEST PAGE 2

7. A.

10 pts

8. A.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **74147**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. The number of independent items rises from 25 to 100. What happens to expected sum and SD of sum?	A. Both quadruple. B. Expected sum stays fixed; SD halves. C. Expected sum quadruples; SD doubles. D. Both halve. 10 pts
2. Why represent a single binomial count by an interval when using a normal approximation?	A. A count always equals its expected value. B. The count can take fractional values. C. The normal distribution has positive probability at every exact point. D. An interval has area under a continuous curve, while a single point has zero area. 10 pts
3. Which quantity describes how sample means vary from sample to sample?	A. The range of all individuals alone. B. The population mean. C. The sample size. D. Standard error. 10 pts
4. One hundred valid 95% intervals are constructed in repeated sampling. Must exactly 95 cover the mean?	A. No; the coverage count itself can vary. B. No; all 100 must cover it. C. Yes; the confidence level fixes that count. D. Yes; any other count proves bias. 10 pts
5. Someone says n=30 guarantees normal sample means for every population. What is the problem?	A. Thirty is too large for any sample. B. Only populations with mean zero qualify. C. Skewness and extreme observations affect how large n needs to be. D. Sample means can never approach normality. 10 pts
6. Population SD is 12 and n=36, with independent observations. sqrt(36)=6. What is the mean's SE?	A. 72. B. 2. C. 6. D. 12. 10 pts
7. A sample has 2 successes and 98 failures. Does a large total alone justify the ordinary normal-approximation proportion interval?	A. No; the observed success count is too sparse. B. Yes; n=100 guarantees every condition. C. No; proportions cannot be estimated at all. D. Yes; failures are numerous. 10 pts

Test continues on the other side of the sheet.

MSC 201 **74147**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. For comparison, sample SD and confidence are held fixed while n increases. What happens to a t mean interval's margin?	A. It becomes the population SD. B. It increases because there are more observations. C. It stays unchanged. D. It decreases as SE and the critical t value decrease. 10 pts
9. A report gives 30 successes among 120 observations. Which is the estimate of population success proportion?	A. 90. B. 120. C. 30. D. 0.25. 10 pts
10. A sample-size formula returns 62.2 observations. Which planned integer size meets that minimum?	A. 63. B. 60. C. 61. D. 62. 10 pts

74147

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4. A.

10 pts

5.

C.

10 pts

6.

B.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

D.

10 pts

9.

D.

10 pts

10. A.

10 pts

MSC 201 **26425**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Why report a confidence interval instead of only a sample mean?	A. It lists every individual observation. B. It expresses uncertainty in estimating the population mean. C. It guarantees the true mean is the midpoint. D. It removes sampling bias. 10 pts
2. Everyone in a few selected clubs is surveyed. What challenges treating all responses as independent?	A. Randomly selecting clubs guarantees individual independence. B. Members of the same club may have correlated responses. C. The sample proportion cannot be computed. D. Club members cannot have binary responses. 10 pts
3. A supplied sampling model gives P(mean<=12)=0.90. What is P(mean>12)?	A. 0.10. B. 1.90. C. 0.50. D. 0.90. 10 pts
4. A proportion estimate is 0.40 with margin 0.05. What interval is this in percentages?	A. 35% to 45%. B. 5% to 40%. C. 39.95% to 40.05%. D. 40% to 50%. 10 pts
5. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. The observations must include negative waits. B. The true sample mean can actually be below every observation. C. The normal model is exact at every tail. D. It is approximation error outside the true support. 10 pts
6. A sample-size formula returns 62.2 observations. Which planned integer size meets that minimum?	A. 61. B. 60. C. 63. D. 62. 10 pts

Test continues on the other side of the sheet.

MSC 201 **26425**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Sample size rises from 25 to 100 under unbiased independent sampling. What happens to the center of sample means?	A. It quadruples. B. It halves. C. It becomes zero. D. It stays at the population mean. 10 pts
8. A sum distribution has mean 100 and SD 4. What total is 2 SD above its mean?	A. 104. B. 102. C. 108. D. 200. 10 pts
9. Population mean is 20; SE of sample means is 2. An observed sample mean is 24. What standardized distance does it have?	A. 12. B. 0.5. C. 2. D. 4. 10 pts
10. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 256. B. 16. C. 15. D. 17. 10 pts

26425

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3. A.

10 pts

4. A.

10 pts

5.

D.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

C.

10 pts

9.

C.

10 pts

10.

C.

10 pts

MSC 201 **89950**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Why report a confidence interval instead of only a sample mean?	A. It expresses uncertainty in estimating the population mean. B. It guarantees the true mean is the midpoint. C. It removes sampling bias. D. It lists every individual observation. 10 pts
2. The plus-four estimate adds two successes and two failures. With 8 successes out of 40, what adjusted fraction is used?	A. 10/40. B. 12/44. C. 8/44. D. 10/44. 10 pts
3. A known-SD mean interval currently has margin 1. To target margin 0.5 at the same confidence, how should n change?	A. Halve n. B. Multiply n by 4. C. Multiply n by 2. D. Add 1 to n. 10 pts
4. Sample SD is 6 minutes and n=9; sqrt(9)=3. What estimated SE is used?	A. 3 minutes squared. B. 6 minutes. C. 2 minutes. D. 18 minutes. 10 pts
5. A sum distribution has mean 100 and SD 4. What total is 2 SD above its mean?	A. 104. B. 108. C. 200. D. 102. 10 pts
6. A question asks whether the combined duration of 40 tasks exceeds one hour. Which variable should be modeled?	A. The count of tasks alone. B. One individual duration. C. The median alone. D. The sum of 40 durations. 10 pts
7. Independent observations come from an exactly normal population. Is n>=30 necessary for their mean to be normal?	A. No; the mean instead becomes uniform. B. Yes; normal means are impossible below 30. C. Yes; individual normality disappears in small samples. D. No; their mean is normal for any positive sample size. 10 pts

Test continues on the other side of the sheet.

MSC 201 **89950**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A symmetric proportion interval is 0.20 to 0.30. What is the estimate and margin?	A. Estimate 0.30; margin 0.25. B. Estimate 0.25; margin 0.05. C. Estimate 0.20; margin 0.10. D. Estimate 0.10; margin 0.05. 10 pts
9. Independent sample size quadruples with population SD fixed. What happens to SE of the mean?	A. It halves. B. It stays unchanged. C. It doubles. D. It quarters. 10 pts
10. Which quantity describes how sample means vary from sample to sample?	A. Standard error. B. The sample size. C. The range of all individuals alone. D. The population mean. 10 pts

89950

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. B.

10 pts

4. C.

10 pts

5. B.

10 pts

6. D.

10 pts

7. D.

10 pts

TEST PAGE 2

8. B.

10 pts

9. A.

10 pts

10. A.

10 pts

MSC 201 **65028**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Sample size rises from 25 to 100 under unbiased independent sampling. What happens to the center of sample means?	A. It becomes zero. B. It stays at the population mean. C. It quadruples. D. It halves. 10 pts
2. From the same sample, a 90% and 99% mean interval are built with the same method. Which statement fits?	A. Both widths must be identical. B. The 90% interval is wider. C. The 99% interval has a different sample mean. D. The 99% interval is wider but has the same center. 10 pts
3. Two samples have the same observed mean above a claimed population mean and the same population SD, but sizes 25 and 100. Which mean is farther above in SE units?	A. The size-100 mean. B. They are equally far. C. The larger sample changes the claimed mean. D. The size-25 mean. 10 pts
4. A symmetric proportion interval is 0.20 to 0.30. What is the estimate and margin?	A. Estimate 0.25; margin 0.05. B. Estimate 0.20; margin 0.10. C. Estimate 0.30; margin 0.25. D. Estimate 0.10; margin 0.05. 10 pts
5. All items in a batch tend to be heavy or light together. What is questionable about sqrt(n) times individual SD for the total?	A. It assumes independent variation. B. It predicts a nonnegative SD. C. It uses the sample size. D. It retains mass units. 10 pts
6. What does a 95% confidence procedure mean in repeated appropriate sampling?	A. 95% of observations lie in every interval. B. Every interval has 95% of its length below the mean. C. About 95% of intervals constructed this way contain the fixed population mean in the long run. D. The sample mean equals the population mean 95% of the time. 10 pts

Test continues on the other side of the sheet.

MSC 201 **65028**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Using the same sample and interval method, confidence increases. What happens to the proportion margin?	A. It increases. B. It stays unchanged. C. The sample proportion becomes the confidence level. D. It decreases. 10 pts
8. Someone says n=30 guarantees normal sample means for every population. What is the problem?	A. Thirty is too large for any sample. B. Skewness and extreme observations affect how large n needs to be. C. Sample means can never approach normality. D. Only populations with mean zero qualify. 10 pts
9. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 256. B. 17. C. 16. D. 15. 10 pts
10. For a normal approximation to the probability of at least a given count, what should happen to the interval representing the cutoff count?	A. Include the entire interval and all intervals above it. B. Use just a single point at the cutoff. C. Exclude the interval and include only those above it. D. Include only the intervals below it. 10 pts

65028

TEST PAGE 1

1.

B.

10 pts

2.

D.

10 pts

3. A.

10 pts

4. A.

10 pts

5. A.

10 pts

6.

C.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

B.

10 pts

9.

D.

10 pts

10. A.

10 pts

MSC 201 **11942**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. At fixed n and confidence, known population SD doubles. What happens to interval width?	A. It stays unchanged. B. It doubles. C. It halves. D. It quadruples. 10 pts
2. All items in a batch tend to be heavy or light together. What is questionable about sqrt(n) times individual SD for the total?	A. It predicts a nonnegative SD. B. It assumes independent variation. C. It uses the sample size. D. It retains mass units. 10 pts
3. Why represent a single binomial count by an interval when using a normal approximation?	A. The normal distribution has positive probability at every exact point. B. The count can take fractional values. C. An interval has area under a continuous curve, while a single point has zero area. D. A count always equals its expected value. 10 pts
4. A report gives 30 successes among 120 observations. Which is the estimate of population success proportion?	A. 30. B. 0.25. C. 120. D. 90. 10 pts
5. A sample of 6 has strong skewness and an extreme outlier. What is sensible about a t interval?	A. Six observations guarantee a normal sampling distribution. B. The outlier must be removed without investigation. C. Its reliability deserves caution. D. It is automatically reliable because t has heavy tails. 10 pts
6. Sample means are centered at 40 with SE 2. Which mean is 1.5 SE above center?	A. 41.5. B. 43. C. 37. D. 42. 10 pts

Test continues on the other side of the sheet.

MSC 201 **11942**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. The course's ordinary normal-approximation interval check requires more than 5 observed successes and more than 5 failures. Which sample passes?	A. 2 successes and 98 failures. B. 5 successes and 95 failures. C. 20 successes and 0 failures. D. 8 successes and 12 failures. 10 pts
8. What does a 95% confidence procedure mean in repeated appropriate sampling?	A. About 95% of intervals constructed this way contain the fixed population mean in the long run. B. 95% of observations lie in every interval. C. Every interval has 95% of its length below the mean. D. The sample mean equals the population mean 95% of the time. 10 pts
9. A population has mean 50. Where are unbiased random-sample means centered?	A. At 50. B. At 50 divided by sample size. C. At zero. D. At the population SD. 10 pts
10. All individual waits lie between 2 and 14. A normal approximation gives a tiny chance that a mean is below 2. How should this be interpreted?	A. The true sample mean can actually be below every observation. B. It is approximation error outside the true support. C. The observations must include negative waits. D. The normal model is exact at every tail. 10 pts

11942

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4.

B.

10 pts

5.

C.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

D.

10 pts

8. A.

10 pts

9. A.

10 pts

10.

B.

10 pts

MSC 201 **81152**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A histogram is made from 500 means, each based on 20 observations. How many values are in that histogram?	A. 20. B. 500. C. 520. D. 10,000. 10 pts
2. A sample grows from 50 to 51 observations. Must its mean move closer to the population mean?	A. No; larger samples always worsen estimates. B. No; individual additions can move it either way. C. Yes; every increase improves the estimate. D. Yes; the law of large numbers guarantees monotone improvement. 10 pts
3. A proportion interval is 20% to 30%. Which conclusion matches the target?	A. The population percentage is estimated between 20% and 30%. B. Exactly 25% of the sample must be successful. C. The mean duration is between 20 and 30 minutes. D. Every individual has a value between 20 and 30. 10 pts
4. The number of independent items rises from 25 to 100. What happens to expected sum and SD of sum?	A. Both quadruple. B. Both halve. C. Expected sum stays fixed; SD halves. D. Expected sum quadruples; SD doubles. 10 pts
5. Population SD is 12 and n=36, with independent observations. sqrt(36)=6. What is the mean's SE?	A. 12. B. 6. C. 72. D. 2. 10 pts
6. An open voluntary poll has 50,000 replies and a very small calculated margin. What remains uncertain?	A. Whether the calculated margin must be negative. B. Whether its respondents represent the target population. C. Whether a proportion is numerical. D. Whether 50,000 exceeds 30. 10 pts

Test continues on the other side of the sheet.

MSC 201 **81152**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A known-SD mean interval currently has margin 1. To target margin 0.5 at the same confidence, how should n change?	A. Multiply n by 2. B. Halve n. C. Add 1 to n. D. Multiply n by 4. 10 pts
8. Sample SD is 6 minutes and n=9; sqrt(9)=3. What estimated SE is used?	A. 3 minutes squared. B. 18 minutes. C. 2 minutes. D. 6 minutes. 10 pts
9. A 95% mean interval is 10 to 14 minutes. Does this say 95% of individual durations lie there?	A. No; it estimates the population mean. B. Yes; the mean and individuals have identical spread. C. Yes; that is the meaning of any confidence interval. D. No; it instead guarantees every duration lies there. 10 pts
10. For a normal approximation to the probability of at most a given count, which intervals belong in the shaded area?	A. Only the intervals above the cutoff count's interval. B. Only the cutoff count's interval. C. The cutoff count's interval and all intervals below it. D. Only the intervals below the cutoff count's interval. 10 pts

81152

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

C.

10 pts

9. A.

10 pts

10.

C.

10 pts

MSC 201 **87194**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Two samples have the same observed mean above a claimed population mean and the same population SD, but sizes 25 and 100. Which mean is farther above in SE units?	A. They are equally far. B. The size-100 mean. C. The size-25 mean. D. The larger sample changes the claimed mean. 10 pts
2. Which t distribution has heavier tails?	A. They have equal tails. B. df=100. C. df=4. D. Both have no tails. 10 pts
3. A 95% confidence interval estimates a population use rate. Does 95% mean that 95% of the population uses the service?	A. Yes; all sample proportions equal 0.95. B. Yes; confidence equals the use rate. C. No; it means 95 people were surveyed. D. No; 95% describes the interval procedure's coverage. 10 pts
4. A question asks whether the combined duration of 40 tasks exceeds one hour. Which variable should be modeled?	A. The median alone. B. The count of tasks alone. C. The sum of 40 durations. D. One individual duration. 10 pts
5. At fixed n and confidence, known population SD doubles. What happens to interval width?	A. It halves. B. It doubles. C. It stays unchanged. D. It quadruples. 10 pts
6. Someone says n=30 guarantees normal sample means for every population. What is the problem?	A. Sample means can never approach normality. B. Thirty is too large for any sample. C. Only populations with mean zero qualify. D. Skewness and extreme observations affect how large n needs to be. 10 pts
7. The number of independent items rises from 25 to 100. What happens to expected sum and SD of sum?	A. Expected sum quadruples; SD doubles. B. Expected sum stays fixed; SD halves. C. Both quadruple. D. Both halve. 10 pts

Test continues on the other side of the sheet.

MSC 201 **87194**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. 400 people are sampled without replacement from 20,000. Does this meet the at-most-10% sample-fraction guideline?	A. Yes; any random sample meets it. B. No; n must be below 30. C. Yes; the fraction is 2%. D. No; the fraction is 20%. 10 pts
9. Which quantity describes how sample means vary from sample to sample?	A. The range of all individuals alone. B. The population mean. C. Standard error. D. The sample size. 10 pts
10. A 95% mean interval is 10 to 14 minutes. Does this say 95% of individual durations lie there?	A. Yes; that is the meaning of any confidence interval. B. No; it instead guarantees every duration lies there. C. Yes; the mean and individuals have identical spread. D. No; it estimates the population mean. 10 pts

87194

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

B.

10 pts

6.

D.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

C.

10 pts

9.

C.

10 pts

10.

D.

10 pts

MSC 201 **73977**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A proportion interval is 20% to 30%. Which conclusion matches the target?	A. The mean duration is between 20 and 30 minutes. B. The population percentage is estimated between 20% and 30%. C. Every individual has a value between 20 and 30. D. Exactly 25% of the sample must be successful. 10 pts
2. A shipment contains 36 packages. Which variable answers the question of average mass per package in that shipment?	A. The largest mass. B. The total mass without dividing. C. The sample mean. D. A single randomly chosen mass. 10 pts
3. Independent observations come from an exactly normal population. Is n>=30 necessary for their mean to be normal?	A. No; their mean is normal for any positive sample size. B. Yes; normal means are impossible below 30. C. Yes; individual normality disappears in small samples. D. No; the mean instead becomes uniform. 10 pts
4. Using the same sample and interval method, confidence increases. What happens to the proportion margin?	A. It increases. B. It stays unchanged. C. The sample proportion becomes the confidence level. D. It decreases. 10 pts
5. Population SD is 12 and n=36, with independent observations. sqrt(36)=6. What is the mean's SE?	A. 12. B. 72. C. 2. D. 6. 10 pts
6. From the same sample, a 90% and 99% mean interval are built with the same method. Which statement fits?	A. Both widths must be identical. B. The 90% interval is wider. C. The 99% interval is wider but has the same center. D. The 99% interval has a different sample mean. 10 pts

Test continues on the other side of the sheet.

MSC 201 **73977**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A mean interval is 10 to 14 minutes. Which report matches its target?	A. The procedure estimates the population mean duration between 10 and 14 minutes. B. Exactly 95 of the next 100 durations will lie there. C. Every future duration will be between 10 and 14. D. The median is guaranteed to equal 12. 10 pts
8. A supplied sampling model gives P(mean<=12)=0.90. What is P(mean>12)?	A. 0.50. B. 0.10. C. 1.90. D. 0.90. 10 pts
9. A small random independent sample comes from a normal population. Is n<30 alone a reason to reject a t interval?	A. Yes; normality is irrelevant below 30. B. No; it must use a proportion interval. C. Yes; t requires at least 30 in every case. D. No; small normal samples can support the t method. 10 pts
10. For n=20, a total exceeding 240 is equivalent to which mean event?	A. Mean exceeding 20. B. Mean below 12. C. Mean exceeding 12. D. Mean exceeding 240. 10 pts

73977

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3. A.

10 pts

4. A.

10 pts

5.

C.

10 pts

6.

C.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

B.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **22143**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. The 95th percentile of a sample-mean distribution is 14. What does it describe?	A. About 95% of repeated sample means are at most 14. B. 95% of all individuals must be at most 14. C. Every sample mean is 14. D. The unknown mean has become 14. 10 pts
2. A one-sample t interval uses n=16. What degrees of freedom are used?	A. 15. B. 16. C. 256. D. 17. 10 pts
3. A 95% confidence interval estimates a population use rate. Does 95% mean that 95% of the population uses the service?	A. Yes; confidence equals the use rate. B. No; 95% describes the interval procedure's coverage. C. Yes; all sample proportions equal 0.95. D. No; it means 95 people were surveyed. 10 pts
4. Everyone in a few selected clubs is surveyed. What challenges treating all responses as independent?	A. Club members cannot have binary responses. B. Members of the same club may have correlated responses. C. Randomly selecting clubs guarantees individual independence. D. The sample proportion cannot be computed. 10 pts
5. A 95% mean interval is 10 to 14 minutes. Does this say 95% of individual durations lie there?	A. No; it instead guarantees every duration lies there. B. No; it estimates the population mean. C. Yes; that is the meaning of any confidence interval. D. Yes; the mean and individuals have identical spread. 10 pts
6. Independent observations come from an exactly normal population. Is n>=30 necessary for their mean to be normal?	A. No; the mean instead becomes uniform. B. Yes; normal means are impossible below 30. C. No; their mean is normal for any positive sample size. D. Yes; individual normality disappears in small samples. 10 pts

Test continues on the other side of the sheet.

MSC 201 **22143**

Unit 4 Test

Name: _____ Date: _____

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Circle one response per question in the right column.

7. Population SD is 12 and n=36, with independent observations. sqrt(36)=6. What is the mean's SE?	A. 72. B. 6. C. 12. D. 2. 10 pts
8. A known-SD mean interval currently has margin 1. To target margin 0.5 at the same confidence, how should n change?	A. Multiply n by 2. B. Halve n. C. Add 1 to n. D. Multiply n by 4. 10 pts
9. Twenty items have sample mean mass 5 g. What is their total mass?	A. 0.25 g. B. 100 g. C. 25 g. D. 5 g. 10 pts
10. Sample size rises from 25 to 100 under unbiased independent sampling. What happens to the center of sample means?	A. It halves. B. It quadruples. C. It becomes zero. D. It stays at the population mean. 10 pts

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TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. B.

10 pts

4. B.

10 pts

5. B.

10 pts

6. C.

10 pts

TEST PAGE 2

7. D.

10 pts

8. D.

10 pts

9. B.

10 pts

10. D.

10 pts

MSC 201 **49478**

Unit 4 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Individual mean mass is 3 g. What is the expected total for 10 items?	A. 10 g. B. 3 g. C. Exactly 30 g in every shipment. D. 30 g. 10 pts
2. Using the same sample and method, confidence rises from 90% to 99%. What happens to interval width?	A. It decreases. B. It stays unchanged. C. It becomes zero. D. It increases. 10 pts
3. Increasing n from 100 to 400 halves SE. Would adding the same 300 observations to any starting n always halve SE?	A. Yes; SE depends only on added observations. B. No; the ratio of sample sizes matters. C. Yes; an extra 300 always halves it. D. No; SE never changes. 10 pts
4. A sample has 2 successes and 98 failures. Does a large total alone justify the ordinary normal-approximation proportion interval?	A. No; the observed success count is too sparse. B. No; proportions cannot be estimated at all. C. Yes; n=100 guarantees every condition. D. Yes; failures are numerous. 10 pts
5. A symmetric interval is 8 to 12. What is the margin of error?	A. 10. B. 4. C. 2. D. 12. 10 pts
6. The CLT supports an approximately normal distribution of large-sample means. What does it not claim?	A. Larger independent samples can have smaller SE. B. Sampling distributions concern statistics. C. Means can have approximately normal sampling distributions. D. Individual observations become normally distributed. 10 pts
7. An exponential population has mean 5. Which is generally more unusual under independent sampling: one duration above 7 or 100 durations averaging above 7?	A. The single duration above 7. B. The average of 100 above 7. C. They have equal probability. D. Neither can occur. 10 pts

Test continues on the other side of the sheet.

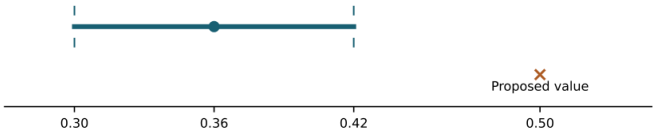
MSC 201 **49478**

Unit 4 Test

Name: _____ Date: _____

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Circle one response per question in the right column.

8. A population proportion interval is 0.30 to 0.42. How should a proposed proportion 0.50 be described?  	A. Proved impossible under every future sample. B. Outside the interval and incompatible with this interval estimate. C. Compatible because 0.50 is a valid probability. D. The required sample estimate. 10 pts
9. A small random independent sample comes from a normal population. Is n<30 alone a reason to reject a t interval?	A. No; it must use a proportion interval. B. Yes; t requires at least 30 in every case. C. No; small normal samples can support the t method. D. Yes; normality is irrelevant below 30. 10 pts
10. A shipment contains 36 packages. Which variable answers the question of average mass per package in that shipment?	A. A single randomly chosen mass. B. The largest mass. C. The sample mean. D. The total mass without dividing. 10 pts

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TEST PAGE 1

1.

D.

10 pts

2.

D.

10 pts

3.

B.

10 pts

4. A.

10 pts

5.

C.

10 pts

6.

D.

10 pts

7.

B.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

C.

10 pts

10.

C.

10 pts